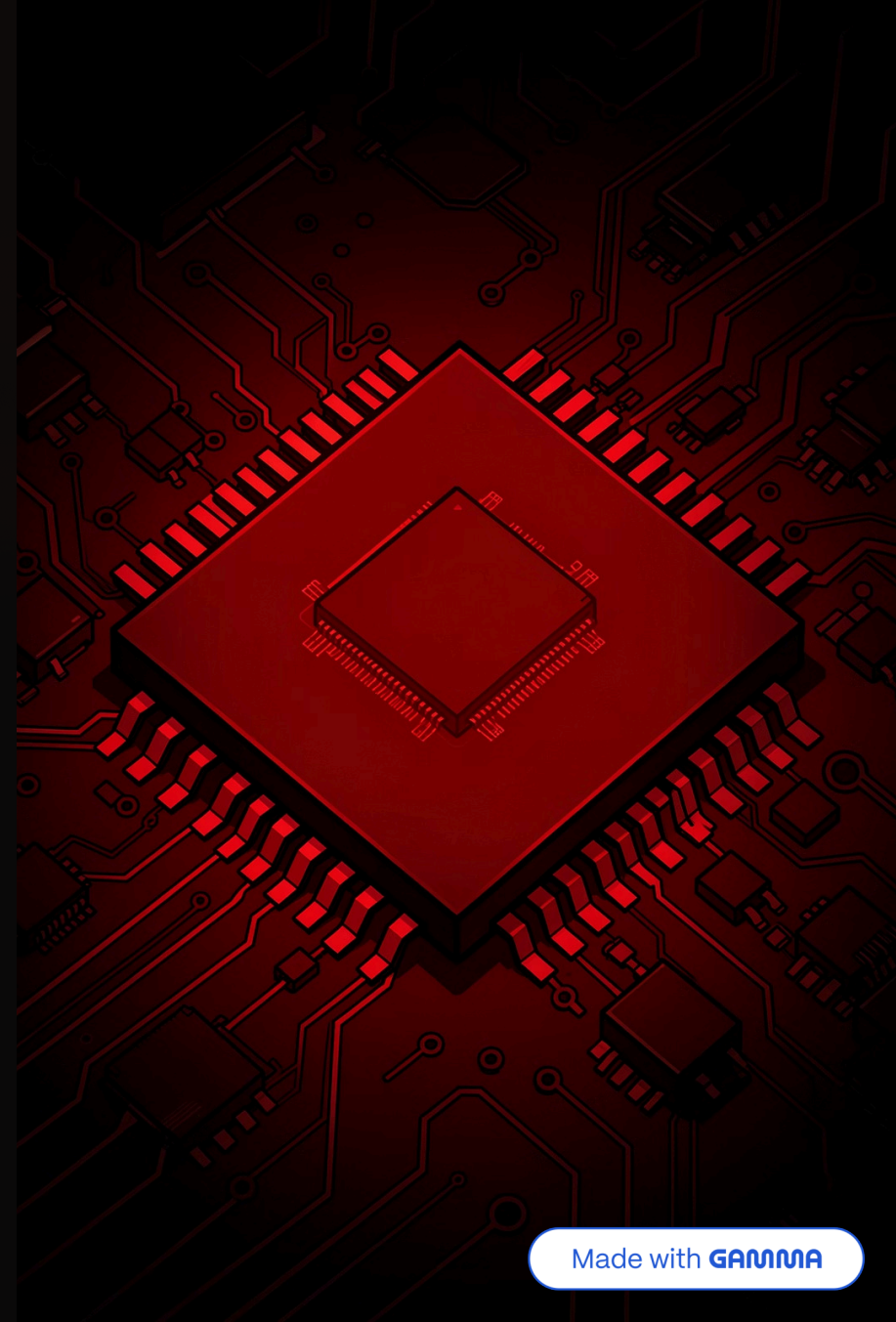


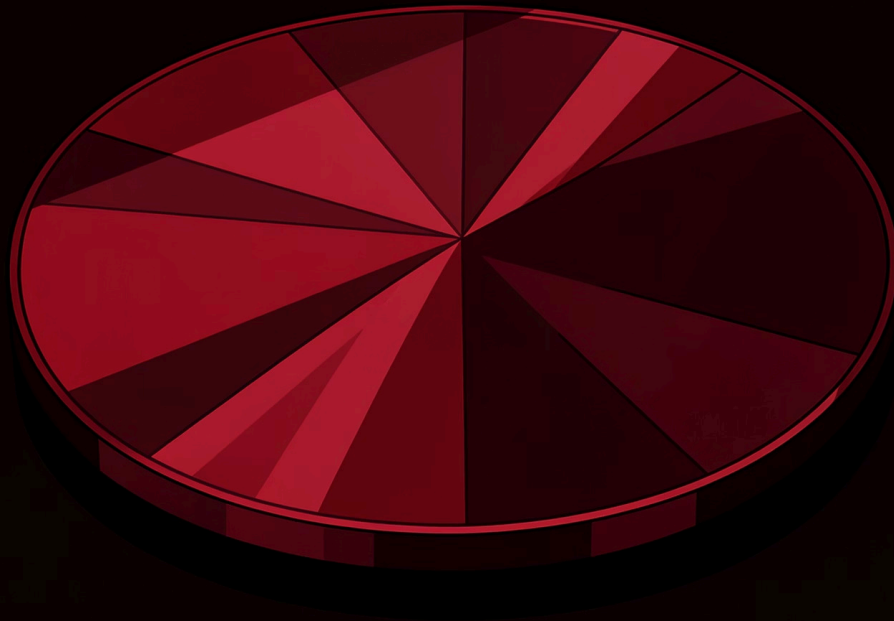
Semiconductor Devices

Powering the Modern World

From the smartphone in your pocket to the satellites orbiting Earth — semiconductors are the invisible engines driving every technology we depend on.



What is a Semiconductor?



A semiconductor is a material whose electrical conductivity falls between a conductor and an insulator. By controlling its properties through **doping** — adding impurities — engineers can precisely regulate how electricity flows.

Silicon (Si)

The most widely used semiconductor material, making up over 95% of all chips produced.

Band Gap

The energy gap between valence and conduction bands determines how easily electrons flow.

From Silicon to Circuits: How Semiconductors Are Made

Chip fabrication is one of the most complex manufacturing processes on Earth — requiring extreme precision across hundreds of steps in ultra-clean environments.

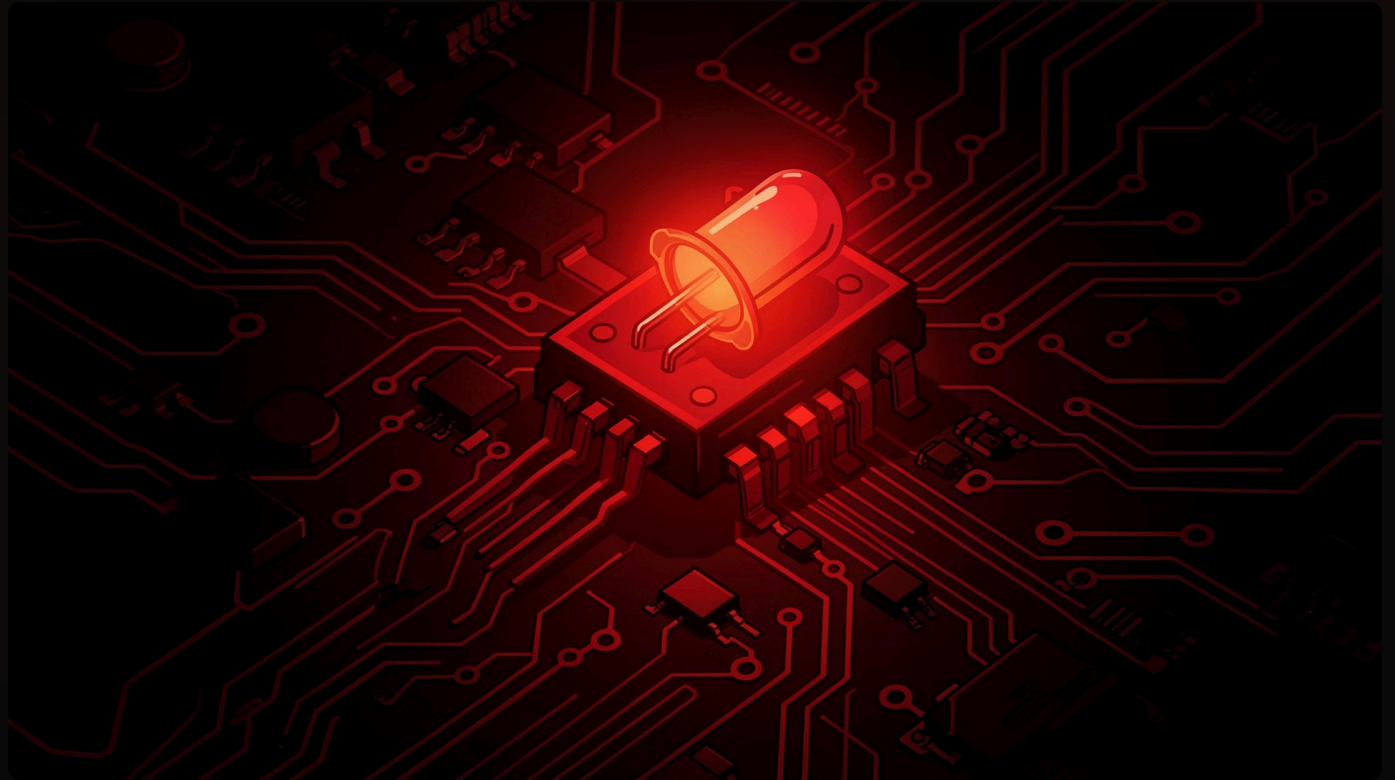


Modern fabs operate at **3nm process nodes**, packing billions of transistors onto a fingernail-sized chip — each layer aligned with nanometer-level accuracy.

Diodes: The One-Way Street for Electricity

How It Works

A diode allows current to flow in only **one direction**. It's formed by joining a P-type and N-type semiconductor, creating a PN junction that blocks reverse current.



Rectification

Converts AC to DC in power supplies

LEDs

Light-emitting diodes for displays & lighting

Zener Diodes

Voltage regulation and protection circuits

Transistors: The Heart of All Modern Electronics



Electronic Switch

A transistor can turn current on or off in nanoseconds — the fundamental operation behind all digital logic.



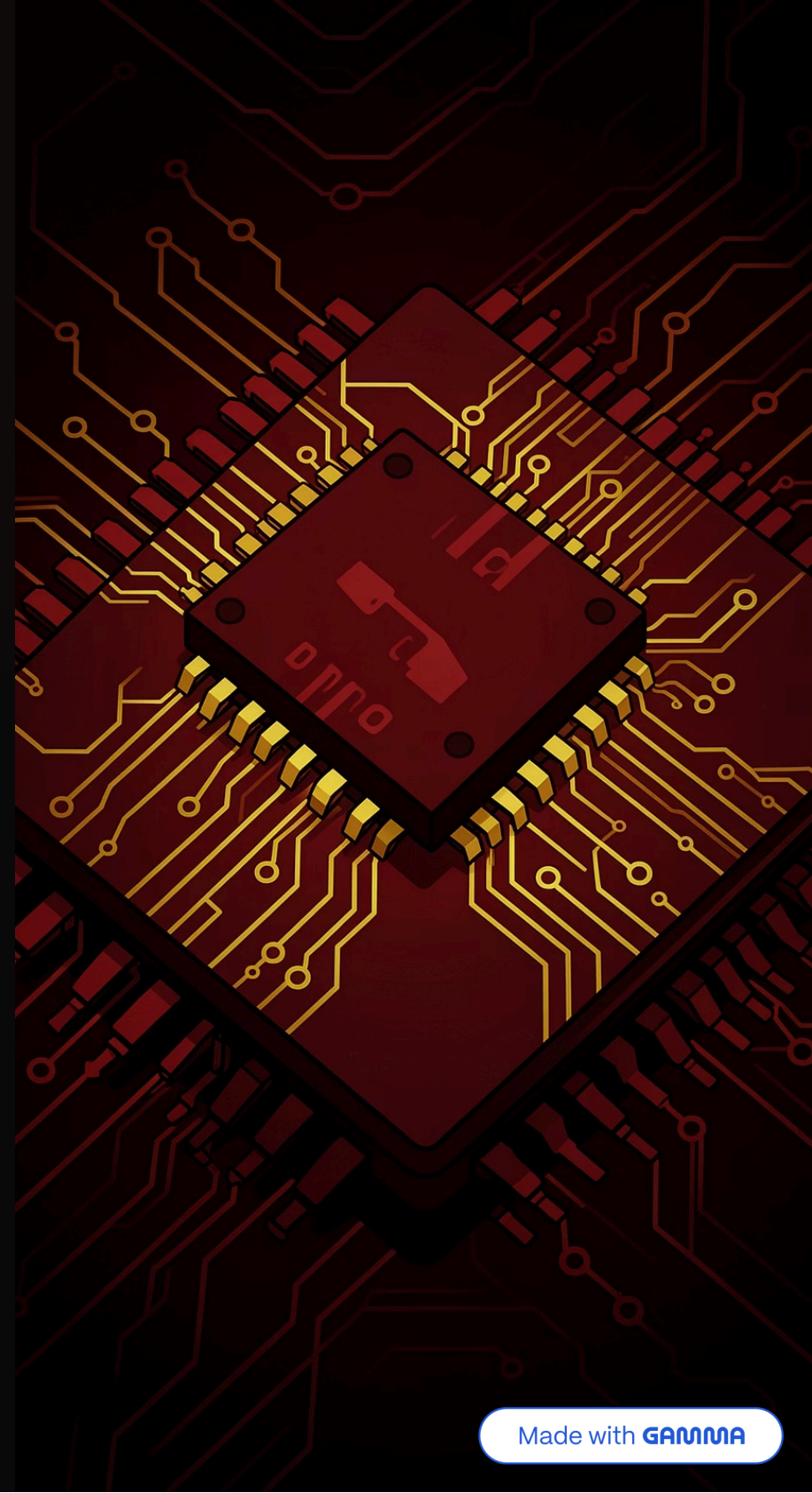
Signal Amplification

Small input signals are amplified into larger outputs, enabling radios, audio systems, and sensors.

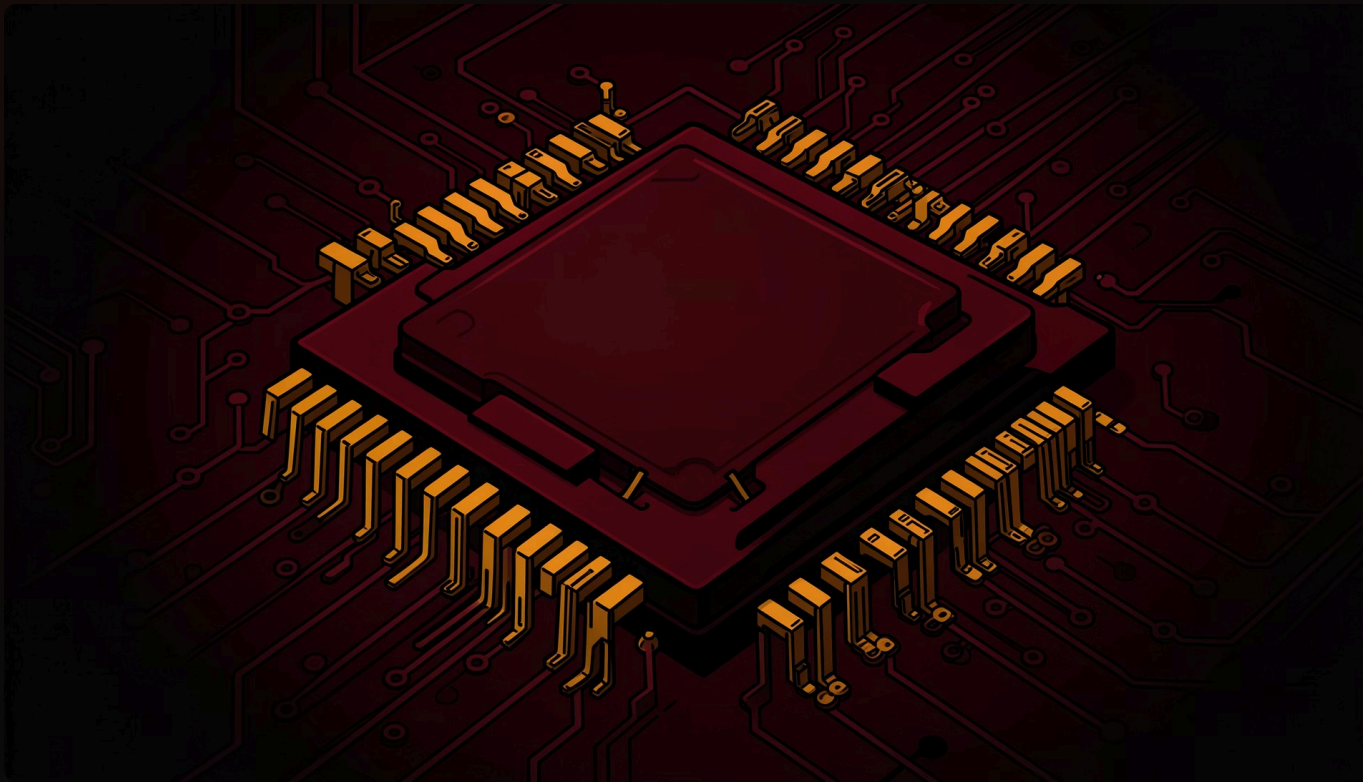


Logic Gates

Billions of transistors combine to form AND, OR, and NOT gates — the building blocks of computation.



Integrated Circuits: Miniaturization & Complexity



An **Integrated Circuit (IC)** packs millions to billions of transistors, resistors, and capacitors onto a single silicon chip. Moore's Law observed that transistor density doubles roughly every two years.

1

SSI / MSI

Few to hundreds of transistors (1960s)

2

LSI / VLSI

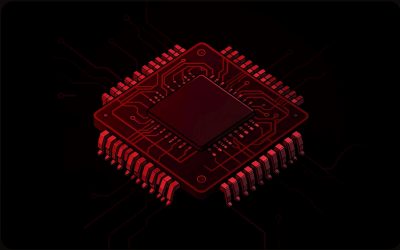
Thousands to millions (1970s–90s)

3

ULSI / SoC

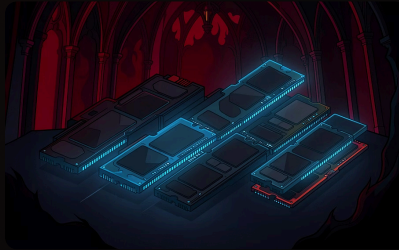
Billions on a single chip today

Applications in Computing: CPUs, Memory & AI Accelerators



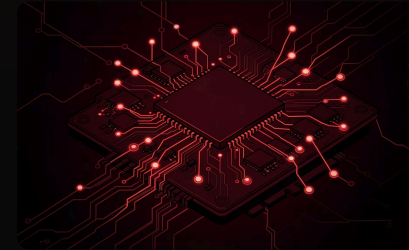
Central Processing Units

Modern CPUs contain **50+ billion transistors**, executing billions of instructions per second across multiple cores.



Memory (DRAM & SRAM)

Volatile memory stores active data. DRAM cells use a single transistor-capacitor pair; SRAM uses six transistors per bit for speed.

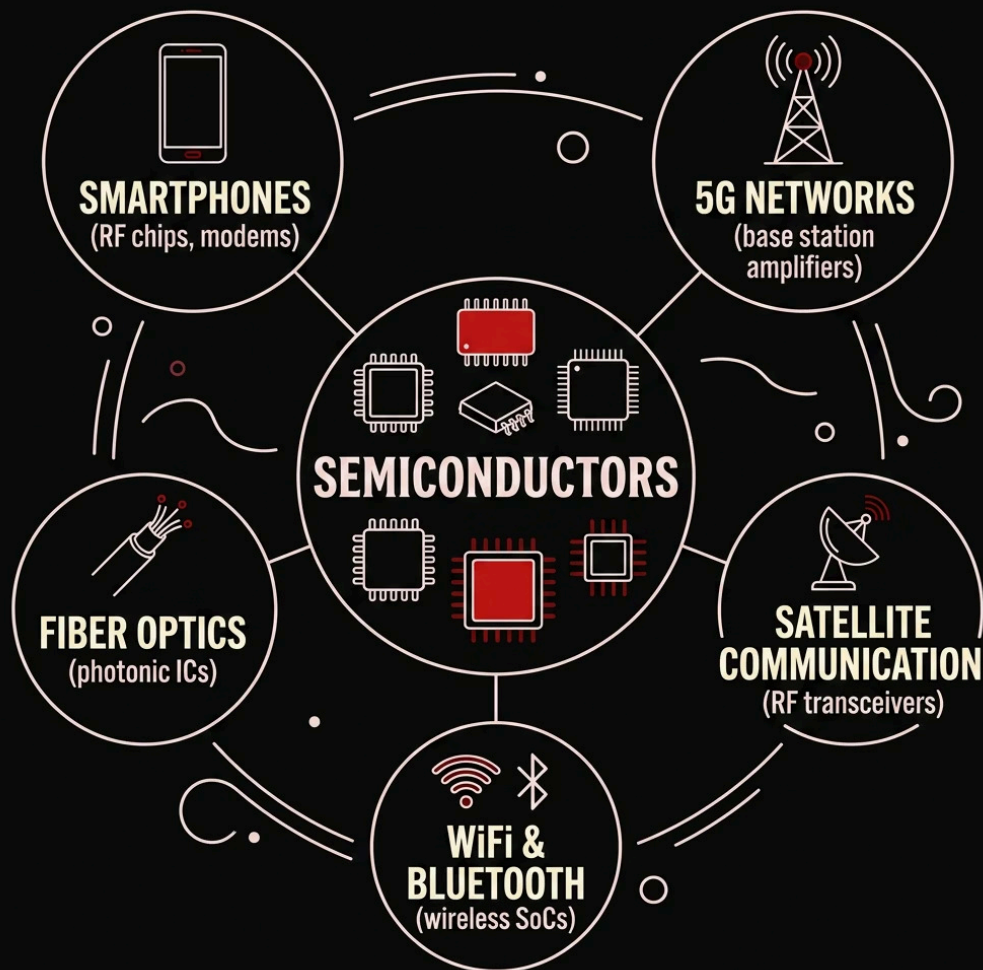


AI Accelerators (GPUs & TPUs)

Specialized chips with thousands of parallel cores optimized for matrix operations powering machine learning and AI.

Semiconductors in Communication

Every call, text, and data packet relies on semiconductor devices to transmit, receive, and process signals across global networks.



Key Enabling Technologies

- **RF Transistors** — Amplify and filter wireless signals
- **Power Amplifiers** — Boost signal strength for long-range transmission
- **Photonic ICs** — Convert electrical signals to light for fiber optics
- **Baseband Processors** — Encode and decode digital data streams

Beyond Electronics: Sensors, Power & Renewable Energy



Sensors & IoT

MEMS sensors, image sensors (CMOS), and environmental sensors enable smart devices, autonomous vehicles, and industrial monitoring.



Power Electronics

Wide-bandgap semiconductors (SiC, GaN) enable efficient power conversion in EVs, inverters, and fast chargers.



Solar & Renewables

Photovoltaic cells convert sunlight directly into electricity. Semiconductor inverters manage grid integration and energy storage.

The Future of Semiconductors

1

2nm & Beyond

Next-generation process nodes pushing atomic-scale manufacturing limits

2

Chiplets & 3D Stacking

Modular chip architectures and vertical integration for performance gains

3

Quantum Computing

Semiconductor-based qubits promising exponential computational power

4

Neuromorphic Chips

Brain-inspired architectures for ultra-efficient AI at the edge

📌 **Key Takeaway:** Semiconductors are not just components — they are the foundation of every major technological leap ahead. Investment in R&D, materials science, and manufacturing capacity will define the next era of innovation.

